

Can Personalized Agents Predict Social Experience?

Research proposal for Professor Michael S. Bernstein

Draft

1 Motivation and prior work

Entering a new community creates opportunities to meet people, but finding meaningful connections remains difficult. We ask whether a personalized AI agent can predict whom someone will feel compatible with, before any real-life interaction. We hypothesize that a brief interview about a person’s context and goals can provide enough information for accurate predictions, and that feedback from real encounters can improve them.

Park et al. show that LLM agents grounded in rich interview and survey self-reports can predict individuals’ responses across survey and behavioral tasks.^[1] We propose extending this idea to interpersonal experience: predicting a person’s own assessment of specific, previously unmet partners.

The Stanford Marriage Pact converts questionnaire responses into numerical vectors, scores compatibility using cosine similarity, and assigns matches across campus while accounting for preferences and dealbreakers. It reports that 3–4% of its matches went on to date for a year or longer.^[2]

We argue that this formulation has a central limitation for predicting individual preferences: it treats similarity as a proxy for attraction. Shared beliefs and values do not necessarily mean two people will click; the qualities someone has need not be the qualities they seek in another person. Cosine similarity between the same two profiles is symmetric, whereas social preferences can be directional and depend on what each person wants now.

Whether two people ultimately become friends or romantic partners depends on how their interactions unfold and on circumstances beyond the initial match. We therefore investigate a more immediate question: whether personalized LLM agents can accurately rank a pool of available people by how worthwhile a first interaction would be for a given individual. These rankings consider that individual’s current context and goals and what each potential partner can offer. Accuracy will be assessed against participants’ own evaluations after real encounters.

2 Proposed approach

We propose a study with newly arrived Stanford students, who are entering a new community and beginning to build friendships, find study partners, and form other meaningful connections. Using a platform we have developed, participants will complete a brief interview about their context and goals, take part in structured, in-person introductions, and provide private feedback after each encounter. This allows us to test whether agents personalized through the interview can predict which encounters participants will find worthwhile, and whether feedback from earlier meetings improves predictions for new partners.

1. **Initial interview.** Participants complete a brief, adaptive text interview about their context and the connection they are seeking. The interview begins with five or six questions that adapt to their answers, providing the initial information for their personalized agent.
2. **Predictions and introductions.** Before participants meet, agents predict how compatible each person will feel with potential partners, given the connection they seek. Predictions are recorded and hidden from participants. The platform considers predictions in both directions when recommending mutually promising introductions.
3. **Conversation.** Participants meet their assigned partner with a common conversation time and opening prompt.

4. **Private feedback.** After each conversation, participants independently answer: “How compatible did you feel with this person, given the connection you were seeking?” on a scale from 1 (*not at all*) to 7 (*extremely*), and briefly explain their rating.
5. **Agent update.** Adaptive agents use earlier feedback and brief follow-up interviews to update their representation of the participant. Subsequent rounds begin at step 2 with a previously unmet partner, allowing us to test whether these updates improve predictions for new people.

The pilot will refine the interview length, number of introductions, and conversation duration.

3 Proposed study and model comparisons

We propose recruiting newly arrived Stanford students for a structured campus gathering, potentially outdoors. A platform we have developed would support the interview, introductions, and feedback collection. Participants would complete up to ten introductions of approximately 10–15 minutes, with the duration and number of sessions refined in a pilot.

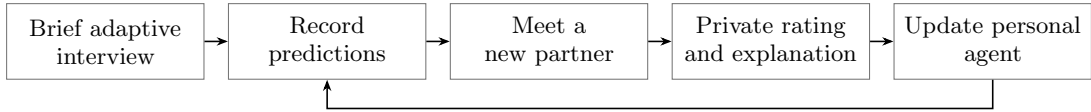


Figure 1. Study procedure. Each subsequent encounter involves a new partner.

The interview targets a few minutes and begins with five or six questions that adapt to the participant’s answers. Before each meeting, predictions are saved and kept hidden from participants. Partners receive the same conversation time and a common opening prompt. Afterwards, each privately answers: “*How compatible did you feel with this person, given the connection you were seeking?*” on a 1–7 scale, from *not at all* to *extremely*, and briefly explains the rating. A fixed update procedure uses earlier feedback to make predictions for new partners.

Each participant completes one sequence of encounters. Four versions of the same underlying LLM predict those same encounters in the background, allowing a direct comparison against identical human experiences. All receive the same initial partner descriptions and the participant’s current goal.

Model	Information about the participant
General	No personal interview or encounter history.
Initially personalized	Initial interview only; the personal representation stays fixed.
Learns from ratings	Initial interview, prior encounter ratings, and brief follow-up interviews.
Learns from ratings and explanations	The same interviews and ratings, plus the participant’s explanations.

Comparing the initially personalized and general models tests the value of the initial interview. The two adaptive versions receive identical follow-up interviews. Their comparison isolates what explanations add; comparison with the fixed agent tests the combined interview-and-feedback update process. For example, a participant may explain that feeling listened to mattered more than shared interests. We test whether this information improves predictions for later partners. A separate cosine-similarity baseline over fixed representations of the initial profiles directly tests whether the agents add predictive value beyond resemblance.

4 Evaluation

The main outcome is agreement with each person’s preference ranking. For a participant who meets ten people, we assess whether the predictions identify those with whom they report the greatest compatibility. Initial rankings are recorded before any meetings. At prespecified feedback checkpoints, each model also ranks a common subsequent set of new partners, before meeting anyone in that set. We compare these fixed rankings with ratings from those later encounters, testing improvement without using an encounter’s own outcome to predict it.

Ranking agreement, accounting for tied ratings, is primary; mean absolute error on the 1–7 ratings is secondary. Comparisons are paired within participants and reported with uncertainty that accounts for repeated participants and partners. Learning curves track performance against feedback count and interview time. Direct agent predictions and the recommendation layer’s estimates are evaluated separately.

We propose including randomly assigned eligible partners alongside model-guided introductions, keeping each participant’s goal fixed within a session. Analyses cover partners they actually meet. A pilot will inform session length, sample size, assignment, and a preregistered analysis with meaningful effect sizes and missing-data handling. Claims about whether the system improves social outcomes would require a separate comparison of assignment policies.

5 Expected contributions and requested supervision

1. **A benchmark for predicting perceived compatibility.** Measure how closely predictions made before people meet recover each participant’s preferences for a stated connection goal.
2. **Evidence about learning from brief feedback.** Establish whether personalization helps, whether subsequent ratings improve predictions for new partners, and whether explanations add value relative to the time participants invest.
3. **A dataset of repeated social encounters.** Link interviews, goals, partner descriptions, assignments, predictions, ratings, and explanations. Consented future reuse could support research on changing preferences and model development, with separate development and held-out evaluation data.

A lack of improvement would identify limits of the proposed representation and update procedure. Findings would concern brief encounters in this population; long-term relationships and generalization to other populations remain untested.

We ask Professor Michael Bernstein to serve as Protocol Director and help us finalize the protocol for Stanford IRB review. Our team would prepare the study materials and operate the platform under his scientific and ethical oversight. Participation would be voluntary, with consent covering personal representations, model processing, and any future data reuse. Ratings and explanations would remain private from conversation partners. Recruitment and data collection would begin only after the required IRB review and authorization.

References

- [1] Park, J. S., et al. (2026 revision). LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals. arXiv:2411.10109v3.
- [2] Marriage Pact. Who We Are. Accessed September 24, 2026.